Devarsh Mahendrabhai Vispute

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PROFESSIONAL SUMMARY

Ph.D. in Mechanical Engineering at Carnegie Mellon with expertise in **thermal analysis, numerical simulation, and mechanical design**. Experienced in academic research with thermal and structural analysis with experimental validation. Passionate about innovative **thermal management solutions** and developing cutting-edge thermal systems to solve real-world problems.

EDUCATION

Carnegie Mellon University, Pittsburgh, PA
Doctor of Philosophy in Mechanical Engineering (GPA: 3.98/4.0) May 2025
Master of Science in Mechanical Engineering (GPA: 4.0/4.0) May 2021
Sardar Vallabhbhai National Institute of Technology, Surat, India
Bachelor of Technology in Mechanical Engineering (GPA: 9.5/10) May 2017

SKILLS

Computational Analysis & Simulation: ANSYS Mechanical, ANSYS Fluent, COMSOL Multiphysics
Programming and Statistical Analysis: MATLAB, ANSYS APDL, Design of Experiments (DOE), Minitab, Python
Modeling and Prototyping: SolidWorks, Laser Cutting, Soldering, 3D Printing, Autodesk Fusion, CNC Routing
Subject Matter Knowledge: Heat Transfer, Thermal Design, Fluid Mechanics, Solid Mechanics, Manufacturing Technologies

EXPERIENCE

Postdoctoral Research Associate, Carnegie Mellon University, Pittsburgh, PA Oct 2025 – Present
- Designing and developing a pressure modulated organ cryopreservation system using novel isochoric preservation technique, consisting of high-pressure chambers, cooling system and data acquisition module (NIH funded)
- Providing technical support and troubleshooting assistance for single-chamber isochoric experimental setup at *University of North Carolina Charlotte, NC*
- Mentoring a PhD student to conduct thermos-fluids simulation for isochoric preservation method using *COMSOL Multiphysics*

Operations Executive, Hindustan Petroleum Corporation Limited, Akola, India Jul 2017 – Jul 2019
- Reduced loading time by 20 minutes, led process optimization and automation initiatives through data-driven workflow improvements
- Conducted root cause analysis and failure investigations of automation and safety systems, ensuring reliability and minimal downtime with preventive maintenance
- Managed capital expenditure projects ($195K+), collaborating with engineers, vendors, and stakeholders to implement engineering solutions for system efficiency, safety, and sustainability, earning the “Smart Terminal Award” in April 2019

THESIS

Title: Thermomechanics Effects and Ice Formation During Organ Cryopreservation Aug 2020-May 2025

Thermomechanical Stress Analysis of Cryopreservation of Human Popliteal Artery
- Reduced thermal stress by 64% by optimizing rewarming process during thermomechanical stress simulation of popliteal artery using *ANSYS*
- Mentored a UG student to create 3D models of blood vessel in cryobag using *SolidWorks*, results presented in a poster presentation by UG student at *Meeting of the Minds, CMU*

Mathematical Modeling of Surface Deformation in CPA During Glass Transition
- Designed and validated computational models (*COMSOL*) for heat transfer and fluid flow in viscoelastic fluid to study surface deformation during cryopreservation
- Reduced simulation time by 78% by simplifying Multiphysics model for thermal contraction and temperature gradient-driven deformation in free surface at cryogenic temperatures

Multiscale Analysis of Physical Effects During Crystallization Using Snomax
- Designed and conducted experimental study to quantify physical effects during crystallization in 12+ cryoprotective agents (CPA) at micro- and macroscale using *Differential Scanning Calorimetry (DSC)* and *Scanning Cryomacroscopy*
- Conducted CPA characterization using *DSC* along with statistical analysis of data on using *Minitab* to highlight significance of using ice nucleating agent during partial freezing

ACADEMIC PROJECTS

Thermal Design and Modeling of Portable Medicine Cooler (Carnegie Mellon University) Jan 2020 - May 2020

- Designed a prototype thermal management system for portable medicine storage (below 4°C), integrating thermal insulation and active cooling to ensure viability of medicine in extreme conditions (team of 4)
- Defined heat transfer performance criteria and optimized component selection to enhance cooling stability and manufacturability at reduced cost and integrated findings from *ANSYS* simulation and testing to refine system design
- Received Best thermal design award among 4 groups working on similar problem with different applications

Study of perfusion of cryoprotective agent (CPA) and its effects in extracellular matrix using microfluidic devices (Carnegie Mellon University) Aug 2023 - Dec 2023

- Simulated CPA perfusion in extracellular matrix (ECM) using COMSOL to demonstrate effect of flow parameters on CPA concentration in ECM
- Developed a research proposal (R03 format) to use 3D microfluidic device along with computational modeling to analyze CPA perfusion in tissues

CFD Analysis of Rotary-bell Atomizer for Paint Shops (Carnegie Mellon University) Feb 2020 - May 2020

- Performed comprehensive CFD analysis using ANSYS Fluent to model paint flow dynamics in a high-speed rotary bell atomizer for automotive paint shops to predict droplet distribution and film thickness
- Increased paint transfer efficiency by 37%, improved film thickness distribution by modifying rotary bell design and optimizing operational parameters in team of 3

Design of Thermal Measurement System for Nanofibers (Carnegie Mellon University) Nov 2019 - May 2020

- Designed an automated thermal characterization system for 1D nano-scale materials operating at cryogenic temperatures (as low as 4 K)
- Developed custom instrumentation integrating a Wheatstone bridge circuit, digital rheostat, and Arduino-based control system to measure thermal conductivity of nanofibers

PUBLICATION

Devarsh Vispute, Yoed Rabin. "Thermomechanics Analysis of Cryopreservation by Vitrification of the Popliteal Artery" *PLOS ONE* (Accepted for publication)

Devarsh Vispute, Simona Baicu, Michael J. Taylor, Yoed Rabin. "Multiscale investigation of Snomax effects on supercooling with application to high-subzero cryopreservation: Calorimetry and cryomacroscopy" *Cryobiology* (2026), https://doi.org/10.1016/j.cryobiol.2026.105650

Devarsh Vispute, Prem Solanki, Yoed Rabin. "Large surface deformation due to thermo-mechanical effects during cryopreservation by vitrification – mathematical model and experimental validation" *PLOS ONE* (2023), https://doi.org/10.1371/journal.pone.0282613

CONFERENCES/PRESENTATIONS

Devarsh Vispute, Simona Baicu, Michael Taylor, Yoed Rabin. "Multiscale Analysis of Ice Formation in 10% DMSO with Snomax" ATP-Bio Annual Meeting, Riverside, CA, 2025. (Poster) ***2nd Prize***

Devarsh Vispute, Yoed Rabin. "Feasibility of preserving structural integrity of popliteal artery during cryopreservation by vitrification" CRYO2023 (Annual meeting of Society of Cryobiology), Minneapolis, MN, 2023. (Presentation)

Devarsh Vispute, Yoed Rabin. "The effects of packaging and specimen handling on thermomechanical stress in popliteal artery during vitrification" ATP-Bio Annual Meeting, Berkeley, CA, 2023. (Poster)

Devarsh Vispute, Prem Solanki, Yoed Rabin. "Experimental validation of a multiphysics mathematical model for body deformation due to thermo-mechanics effects during cryopreservation by vitrification" CRYO2022 (Annual meeting of Society of Cryobiology), Dublin, Ireland - virtual, 2022. (Poster)

LEADERSHIP

Co-Chair, ATP-Bio Scholars Leadership Council (Multi-Institutional ERC, NSF funded) Mar 2023 – Jun 2024

- Facilitated development of data sharing guidelines and a resource database to promote cross-institutional collaboration among 90 scholars
- Modified SLC structure for better alignment with ERC vision and mission, while voicing SLC requirements to management

Session Leader, National Biomechanics Day 2023, Carnegie Mellon University Oct 2022 – Apr 2023

- Presented challenges of thermomechanical stress in organ cryopreservation to 20+ high school students using a scanning cryomacroscope as part of an outreach initiative